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# Curriculum Vitae

## Education

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| September 2025 – ongoing | Carnegie Mellon University, Pittsburgh, PA, United States<br>Walter Noll Research Fellow, Postdoc mathematics |
| 2022 – 2025 | Max Planck Institute for Mathematics in the Sciences, Leipzig, Germany<br>Ph.D. mathematics<br>PhD thesis: "Heat Flow, Mean Curvature Flow and Their Related Schemes",<br>advisor: Prof. Felix Otto |
| 2020 – 2022 | University of Bonn, Germany<br>Master-studies mathematics (grade 1.0, excellent)<br>Master's thesis: "Thresholding for generic mean curvature flow",<br>advisor: Prof. Tim Laux |
| 2017 – 2020 | University of Bonn, Germany<br>Bachelor-studies mathematics (grade 1.0, excellent)<br>Bachelor's thesis: "Starke isoperimetrische Ungleichung" (Sharp isoperimetric inequality),<br>advisor: Prof. Herbert Koch<br>awarded bachelor prize of the BMG (Bonn Mathematical Society) |
| 2009 — 2017 | Georg-Cantor-Gymnasium, Halle, Germany<br>graduation: Abitur (grade 1.0, excellent) |

## Research interests

Geometric flows

Calculus of variations

Parabolic partial differential equations

## Publications and paper

Peer-Reviewed

2026 "Equivalence of weak solution concepts for mean curvature flow", with Tim Laux, Proc. AMS arXiv:2510.16478

2024 "Generic level sets in mean curvature flow are BV solutions", with Tim Laux, J. Geom. Anal. 34, 375 (2024). https://doi.org/10.1007/s12220-024-01819-y

Preprints

upcoming "Learning Manifolds from Discrete Data", with Dejan Slepcev, Lin Chen

upcoming "The Kinetic Undercooling Limit for the One Dimensional Supercooled Stefan Problem and a Natural Solution", with Irene Fonseca, Giovanni Leoni, Mykhaylo Shkolnikov

2024 "Median filter method for mean curvature flow using a random Jacobi algorithm", with Tim Laux, arXiv preprint arXiv:2410.07776
https://mathrepo.mis.mpg.de/Medianfilter/

2024 "The heat flow on glued spaces with varying dimension", arXiv preprint arXiv:2406.09996

## Research stays

11/2025 University of Michigan, United States, visiting Selim Esedoḡlu

07/2025 Heidelberg University, Germany, visiting Tim Laux

04/2024 Regensburg University, Germany, visiting Tim Laux

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| 04/2023 | Heidelberg University, Germany, visiting Tim Laux |

## Contributed Talks & Poster presentation

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| 11/2025 | Workshop “Mathematical Analysis in Machine Learning”, Poster, University of Pittsburgh, Pittsburgh, United States |
| 11/2025 | “AIM Seminar” University of Michigan, Michigan, United States |
| 09/2025 | “CNA seminar” Carnegie Mellon University, Pittsburgh, United States |
| 07/2025 | “Analysis AG seminar” Heidelberg University, Heidelberg, Germany |
| 07/2025 | Summer School “Geometric Analysis and PDEs”, Poster, Chęciny, Poland |
| 10/2024 | “Dresden-Leipzig AG seminar day”, Leipzig, Germany |
| 08/2024 | Conference “Free Boundary Problems: Theory and Application” — "Interfaces and Free Boundaries in Machine Learning", João Pessoa City, Brazil |
| 10/2023 | “Regiomontanus Seminar”, Leipzig, Germany |
| 07/2023 | Summer School “Geometric Analysis and PDEs”, Poster, Chęciny, Poland |
| 07/2023 | “Scientific retreat of the International Max Planck Research School (IMPRS)”, Wittenberg, Germany |
| 11/2022 | “Scientific retreat of the IMPRS”, Trebsen, Germany |

## Organization

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| 10/2024 – 08/2025 | Organization FLINTAQ*-lunches |
| 09/2024 | Organization of Conference “IMPRS Combo III: Cosmology and Quantum Information Theory”, intended for all Ph.D. candidates in mathematics of the IMPRS, Leipzig new concepts: ethics lectures combined with a discussion |
| 10/2023 – 04/2025 | Ph.D. representative for the Max Planck Institute for Mathematics in the Sciences |

## Teaching experience

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| Fall 2026 | Teaching 21-355 Principles of Real Analysis I |
| Summer 2026 | Supervised Reading "Differential Geometry and the Whitney Embedding Theorem" |
| Spring 2026 | Teaching 21-356 Principles of Real Analysis II |
| 10/2024 – 12/2025 | Supervision of a high-school student for the creation of a scientific work |
| 2018 – 2021 | Teaching assistant for: |
| | "Linear algebra 1", "Linear algebra 2", "Analysis 3", "Introduction to robotics", "Introduction to partial differential equations" and "Functional analysis and PDE" |

## Special accomplishments

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| 2020 – 2022 | Studienstiftung des deutschen Volkes, Scholarship of the German Academic Scholarship Foundation; awarded to approximately 0.25% of the German university student population |
| 2015 — 2017 | Besondere Lernleistung (special academic accomplishment, a voluntary research project)<br>"2- und 3-dimensionale Gleichdicke" (curves and surfaces of constant width) |
| 2007 — 2017 | Participation in olympiads in mathematics, physics, chemistry, computer science and latin, for instance: |
| | International physics-olympiad, 3rd round |
| | International chemistry-olympiad, 4th round |
| | National computer science competition, 2nd round |
| | Certamen Franckianum, 2nd round |
| | National mathematics competition, 2nd round |
| | Mathematics Olympiad, up to the selection exams for the International Mathematics Olympics |

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| 2014 | Scientific-practical-work<br>"Erstellung virtueller Welten mit Fraktalen" (Creation of virtual worlds using fractals) |

## Attended Conferences and Workshops

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| 05/2026 | "Perspectives in Applied Mathematics" in Honor of Felix Otto, Leipzig, Germany |
| 10/2024 | "Dresden-Leipzig AG seminar day", Leipzig, Germany |
| 08/2024 | Conference "Free Boundary Problems: Theory and Application", João Pessoa City, Brazil |
| 05/2024 | "IMPRS Combo II: Numerics in Algebra & Analysis", Leipzig, Germany |
| 05/2024 | "n-Cities Seminar", Leipzig, Germany |
| 04/2024 | Workshop on "Calculus of Variations and Partial Differential Equations", Regensburg, Germany |
| 03/2024 | "Scientific retreat of the IMPRS", Halle, Germany |
| 02/2024 | "Applications of Optimal Transportation", Oberwolfach, Germany |
| 11/2023 | "Workshop on Geometry and Machine Learning", Leipzig, Germany |
| 11/2023 | "IMPRS Combo I", Leipzig, Germany |
| 07/2023 | "Unfinished Business: Problems in Applied Geometric Measure Theory, Many-particle Systems, Epidemiology, and Phase Transitions", Bingen, Germany |
| 07/2023 | "Scientific retreat of the IMPRS", Wittenberg, Germany |
| 07/2023 | "Summer School on Geometric Analysis and PDEs", Chęciny, Poland |
| 06/2023 | "Calculus of Variations and Geometric Measure Theory", Pisa, Italy |
| 05/2023 | MPI MiS Summer School on "PDEs and Randomness", Leipzig, Germany |
| 03/2023 | Spring School "Recent trends in the mathematics of complex materials", Bonn, Germany |

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| 12/2022 | "Théorie de la mesure géométrique et calcul des variations", Nancy, France |
| 11/2022 | "Scientific retreat of the IMPRS", Trebsen, Germany |
| 08/2022 | Hausdorff School: "Geometric Analysis", Bonn, Germany |
| 11/2021 | "Workshop on New Trends in Geometric PDEs", Münster, Germany |
| 08/2021 | Workshop on "Geometric and Applied Analysis", Hausdorff Center for Mathematics, Bonn, Germany |

## Special knowledge

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| Languages | German: native<br>English: C1.1<br>Latin: advanced Latin certificate<br>French: A1.1 |
| Programming languages | C: course work<br>C#: private projects, game development<br>Python: private projects, course work, research projects<br>Matlab: course work, research projects<br>Rust: research projects<br>js, css, html: website design |

## Hobbies and interests

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| | Literature<br>Programming & making games<br>Painting<br>Teaching for olympiad preparation<br>Board games |
| association membership | eLeMeNTe e.V. (math association to organize olympiads and math camps)<br>QED e.V. (mathematics organization centered in Bavaria)<br>CDE e.V. (organizes camps and events for people who attended "Deutsche Schüler-Akademie") |

Pittsburgh, 25. December 2025